a Repeat the original forecasting procedure on shuffled outcomes

1 Shuffle outcomes

Within each patient; preserve the chosen patient or block event counts.

2 Rebuild the rule

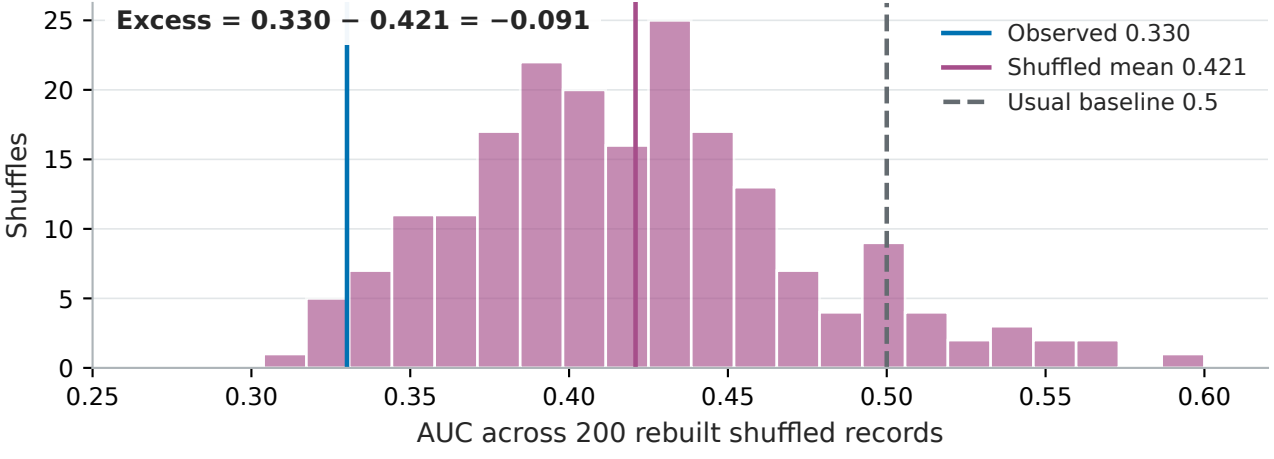
Recalculate outcome-derived inputs; repeat fitting and reference construction.

3 Score each rebuilt record

Use the same patient-level measure. Repeat the shuffle and rebuild.

4 Average, then subtract

Baseline = mean defined shuffled score.
Excess = observed score − baseline.

Keep run-in outcomes and independently learned quantities fixed.

b Worked example: the baseline is computed, not assumed